INCH-POUND
MIL-DTL-14610H
August 15, 2000
SUPERSEDING
MIL-C-14610G
21 April 1983

DETAIL SPECIFICATION

COVERALLS, EXPLOSIVE HANDLER'S

1. SCOPE

1.1 Scope. This detail specification covers the requirements for one type of an explosive handlers coverall intended for wear by personnel engaged in operations involving the handling of ammunition or explosives.

1.2 CLASSIFICATIONS: The coveralls will be of one type in the following sizes:

X-Small	Small	Medium	Large	X-Large
regular	short	short	short	short
	regular	regular	regular	regular
	long	long	long	long

Beneficial comments, (recommendations, additions, deletions) and any data which may be of use in approving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-COCT, 700 Robbins Avenue, Philadelphia, PA 19111-5096 by using the self addressed Standardization Document improvement proposal (DD form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC 8415**

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2. **APPLICABLE DOCUMENTS**

2.1 General. The documents listed in this section are specified in sections 3 and 4. This section does not include documents cited in other sections of this specification or recommended for additional information or an example. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government Documents

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specification and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

A-A-55126 Fastener Tapes, Hook and Loop Synthetic

CODE OF FEDERAL REGULATIONS

(Applications for copies of referenced documents should be addressed to U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328.)

CONTACT ALLERGY: “PREDICTIVE TESTING IN HUMANS”, ADVANCES IN MODERN TOXICOLOGY

(Applications for copies of referenced documents should be addressed to U.S. Army Center for Health Promotion and Preventative Medicine, ATTN: MCHB-DC-TTE, Bldg. E-2100, Aberdeen Proving Grounds, MD 21010-5442.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DoDISS cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) PUBLICATION

ANSI/ASQC Z1.4 Sampling Procedures and Tables for Inspection of Attributes.

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York, NY 10036-8002)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) PUBLICATION

ASTM-D3776 Standard Test Methods for Mass Per Unit Area (Weight) of Woven Fabric

ASTM-D5034 Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Grab test)

ASTM-D737 Standard Test Method of Air Permeability of Textile Fabrics

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ASTM-D6413 Standard Test Methods for Flame Resistance of Textiles (Vertical Test)

ASTM-D1424 Tearing Strength of Woven Fabrics by Falling-Pendulum Type (Elmendorf) Apparatus

ASTM-D2256 Tensile Properties of Yarns by the Single-Strand Method

(Application for copies should be addressed to American Society for Testing and Materials, 100 Barr Harbour Drive, West Conshohocken, PA 19428)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC) PUBLICATION

AATCC-96 Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool

AATCC-81 pH of Water-Extract from Bleached Textiles

AATCC-70 Water Repellency: Tumble Jar Dynamic Absorption Test

(Applications for copies should be addressed to American Association of Textile Chemists and Colorists (AATCC) P.O. Box 12215, Triangle Park, NC 27709-2215)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

3.1 Design and construction. The coveralls shall have a topstitched collar, long sleeves with hook and loop adjustment tabs, a lattice hip pocket on the right back side as worn, a hanger loop with a 2 to 3 inch free opening and side pass throughs. The front opening and side pass throughs shall have hook and loop fastener tape extending down the entire opening for quick removal.

3.1.1 Lattice pocket. Tools weighing up to 8 pounds such as hammers and wrenches shall be able to hang from the open structure pocket. The pocket shall finish 7 to 7-1/4 inches wide and 7-1/2 to 7-3/4 inches long with the raw edges are turned under and stitched. The lattice strips shall measure 5/8 to 1 inch wide.

3.1.2 Seams and stitching. All seams shall be 10-11 stitches per inch for strength. All edges shall be overedged or otherwise clean finished. All hems shall be clean finished with raw edges turned in.

3.1.3 Bartacking. A 1 inch bartack, containing a minimum of 42 stitches shall be located 1/4 $\pm$ 1/16 inch below the slide fastener tape across the crotch seam (see figure). 1/2 horizontal bartacks shall be located at the crossings of each vertical and horizontal strip on the lattice pocket.

3.2 Basic material. The fibers shall be made from cotton which has been spun into singles yarn. The weave for the material shall be a five harness sateen cloth. The use of substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The filling effect side of the natural color cloth shall be the face side. The material shall meet the following requirements in Table I:

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TABLE I. Physical requirements

Characteristic	Minimum	Maximum
Weight (oz per sq. yd)	-	10.5
Breaking strength, lbs.		
Warp	110	-
Filling	100	-
Dimensional change	-	shall be 2% max. in either the warp or filling direction
Air permeability (CU FT/MIN/SQ.)	10.0	-
Flame retardancy:		
Initial after flame time (seconds)	-	2
percent consumed	-	40%
Flame retardancy after 15 launderings:		
the initial after flame time (seconds)	-	2
percent consumed	-	40%
pH	5.0	8.5
Tear Strength (warp and fill)	6.0	-
Dynamic absorption	40%	-

3.3 Toxicity. The flame retardant treated cloth shall show no toxicity to the skin or eyes when used as intended when tested in accordance with 4.3.4.

3.4 Thread. The thread shall be cotton thread or cotton-covered, polyester core thread. The color of the thread shall approximate the color of the basic fabric. The minimum breaking strength shall be 1.8 pounds when tested in accordance with paragraph 4.3.3.

3.5 Hook and loop fastener tape. The hook and loop fastener tape shall be a minimum of 1-inch wide, white in color, and conform to Type II, Class 1 of A-A-55126.

3.6 Labels. Each coverall shall have a commercial-type size label, identification label and instruction label. The identification label shall contain the item description, contractors name, contract number, National Stock Number (NSN) and fiber and laundering information. The identification and instruction label can be combined. The labels shall show durability and colorfastness to laundering. The labels shall be sewn on all four sides to the inside center back, below the collar. The size label shall be positioned above the identification/instruction label. The instruction label shall contain the following inscription:

Wear as outer garment
Keep front closed to assure
body protection.
Keep sleeves adjusted.
Keep pocket clean and free

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of dust accumulation.
Wash at low temperature. Dry thoroughly.
Warning: Upon removal of garment, brush off accumulated foreign matter thoroughly.

3.7 Barcode. Each item shall be individually bar-coded with a tag for personal clothing items. The tags shall be attached to each item, and shall be clearly legible and readable by a scanner. The bar coding element shall be a 13 digit national stock number (NSN). The bar code type shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified.

3.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.9 Patterns. The Government shall furnish a complete set of patterns or a master pattern with grade rules. The government patterns shall be used to create the contractor's working patterns. Minor modifications are permitted to accommodate manufacturing procedures, however, the design and finished measurements must be maintained.

3.9.1 List of pattern parts. Table II is provided to insure that the pattern set is complete.

TABLE II. List of pattern parts

Nomenclature	Cut pieces
Top front	2
Top back 1/	1
Trouser front	2
Trouser Front extension (side entry)	2
Trouser back	2
Right front facing	1
Sleeve	2
Collar 2/	2

1/ The back may be cut in two pieces (1/2 inch seam allowance shall be added for a double-lapped, double-stitched seam.)
2/ The collar may be cut in one piece (eliminate bottom edge seam allowance).

3.9.2 Finished measurements. The finished coveralls shall conform to measurements as shown in Table III.

TABLE III. Finished measurements (inches)

Size	X-Small	Small	Medium	Large	X-Large
½ Chest 1/	20-1/2	22-1/2	24-1/2	26-1/2	28-1/2
½ Waist 2/	17-1/2	19-1/2	21-1/2	23-1/2	25-1/2

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TABLE III. Finished measurements (continued)

Size	X-Small	Small	Medium	Large	X-Large
Sleeve inseams 3/					
Short	-	22	22	22	22
Regular	23	23	23	23	23
Long	-	24	24	24	24
Leg inseams 4/					
Short	-	30	30	30	30
Regular	32	32	32	32	32
Long	-	34	34	34	34
Overall Length 5/					
Short	-	62-1/2	63-1/2	64-1/2	65-1/2
Regular	64-1/2	65-1/2	66-1/2	67-1/2	68-1/2
Long	-	68-1/2	69-1/2	70-1/2	71-1/2

NOTE: Tolerances shall be ± 1/2 inch.

1/ Measure across the front at the base of armhole from folded edge to folded edge.

2/ Measure across the waist from folded edge to folded edge.

3/ Measure from the base of armhole along inseam to bottom of sleeve hem.

4/ Measure from crotch seam along inseam to bottom of leg hem.

5/ Fold coverall in half (back to back) and measure from base of neck (close to hanger loop) straight down to the bottom of the leg.

3.10 Workmanship. The finished coveralls shall conform to the quality of product established by this document.

3.11 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

4. **VERIFICATION**

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.2)
2. Conformance inspection (see 4.3)

4.2 First article inspection. The first article when applicable, submitted in accordance with 3.11, shall be inspected for compliance with design, configuration, workmanship and dimensional requirements. The presence of any defect (see 4.3.5) or failure to pass any test shall be cause for rejection of the first article.

4.3 Conformance inspection. Sampling for visual and dimensional inspection shall be performed in accordance with ANSI/ASQC Z1.4, except where otherwise indicated.

4.3.1 Component and Material Inspection. In accordance with 4.1, components and materials shall be tested in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, qualified in this document or applicable procurement documents.

4.3.2 Test methods. The finished basic material shall be tested for the characteristics listed in Table IV.

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TABLE IV. Material tests

Characteristic	Requirement Table	Test method
Weight	Table I	ASTM-D3776
Breaking strength	Table I	ASTM-D5034
Dimensional change	Table I	ATCC 96 Opt 1C 1/ 2/ 3/
Air permeability (min.)	Table I	ASTM-D737
Flame retardancy:		
Initial after flame time (max.)	Table I	ASTM-D-6413
percent consumed (max.)	Table I	"
After 15 launderings	Table I	AATCC-96, Test Vi, A
After flame time (max.)	Table I	ASTM-D-6413
percent consumed (max.)	Table I	"
pH	Table I	AATCC-81
Tear Strength	Table I	ASTM-D1424
Dynamic Absorption	Table I	AATCC-70

1/ Sample size shall be 22 by 22 inches with 18 inch benchmarks, 6 inches apart.
2/ Use Drying procedure (D) or samples may be dried using a thermostatically controlled circulating oven 221^0 - 230^0 F until dried. The sample shall be allowed to cool for 5 minutes, sprinkled with water to permit damp pressing, and allowed to stand in this condition for 5 minutes.
3/ The dampened specimen shall be pressed with a flatbed press as described in AATCC 96 Section 5.2 or a hand iron as described in AATCC 96 Section 5.10 at a temperature of 275^0 to 300^0 F. Section 7.4 of AATCC 96 shall not apply.

4.3.3 Thread breaking strength. The breaking strength shall be tested in accordance with ASTM D2256.

4.3.4 Toxicity. The cloth shall be tested for toxicity as follows:

a) Title 40, code of Federal Regulations, 1994 Edition;
Section 798.4100 - Dermal Sensitization
Section 798.4470 - Primary Dermal Irritation
Section 798.4500 - Primary Eye Irritation

Marzulli, F & H. Maibach, "Contact Allergy: Predictive testing in Humans". Advances in Modern Toxicology, Volume 4, pp 353-372, 1977.

b) **As an alternative** to animal and human testing, the contractor may provide information which certifies that the cloth is composed of Chemicals and/or materials which have been safely used commercially where prolonged skin contact has occurred.

4.3.5 Visual examination. Coveralls shall be examined for the defects listed in Table V.

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TABLE V. Defects

Defects	Major	Minor
Material defects		
Outside-Any hole, cut, tear, mend, burn, smash, spot or stain, fabric cut on wrong side, stamping marks, color not as specified.	101	
Inside- Defects listed above not affecting performance, serviceability and Appearance.		201
Cutting and marking		
-Any component part not cut in accordance with patterns	102	
-Corrosive metal fastening device or sewn on ticket	103	
Seams and stitching		
-Any component not properly stitched; missing components; components not as specified.	104	
-End of stitching not caught in other seams or not securely backstitched	105	
-Stitch breakage due to tight tension	106	
-Thread breaks not securely overstitched for a 1/2 inch distance		202
-Seams which are twisted, puckered and pleated	107	
-Open seam more than 1 inch	108	
-Open seam less than 1 inch		203
-Any loose, missing or misplaced bartacks	109	
-Thread ends not trimmed or removed		204
-Gage of stitching or stitches per inch irregular	110	
-Loose tension, resulting in loose seam	111	
Finished dimensions		
-Measurements not as specified	112	
Labels		
-Label missing, incorrect, or illegible;	113	
-Bar code not visible on folded, packaged item.		205

5. **PACKAGING**.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materials is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity within the Military Departments Systems Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. The coveralls are intended for use by personnel engaged in operations involving the handling of ammunition or explosives.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this document
b. Issue of DODISS to be cited in the solicitation
c. When first article is required (see 4.2 and 6.3)
d. Packaging Requirements (see 5.1)

6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article will be a preproduction sample. The contracting officer will specify the appropriate type of first article and the number of units to be furnished. The contracting officer will include specific instruction in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Clothing
Pyrotechnics

Custodians:

Army –GL
Navy – NU
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY

GSA-FSS

Review activities:

Army – MD
Navy – AS
Air Force – 82, 11. 6
Navy – CG

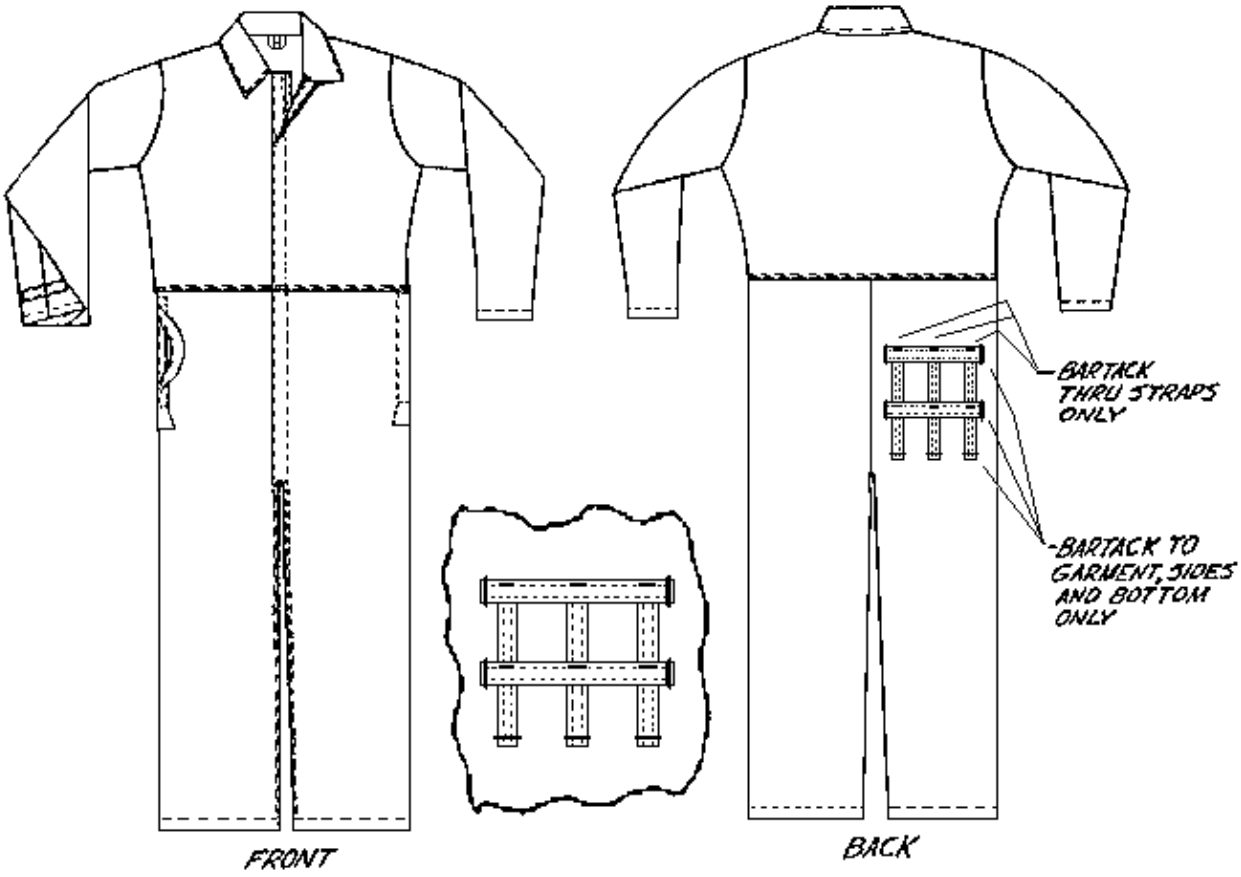
Preparing activity:

DLA-CT

project number: (8415-0156)

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Figure I. Explosive Handler's Coverall



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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7 and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-14610H	**2. DOCUMENT DATE** ***(YYYYMMDD)*** 2000/08/15

3. DOCUMENT TITLE

Coveralls, Explosive Handler's

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER PHILADELPHIA DSCP-C	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN (215) 737- 444-
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave (Bldg 6, C&T) PHILADELPHIA, PA 19111-5092	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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